MTH636 – Game Theory

Lecture 16: Extensive Form Games & Perfect Information

1 Directed Rooted Trees

To formally define an extensive form game, we first need to define the underlying graph structure.

Definition: Directed Rooted Tree

A **directed rooted tree** is a triplet $T = (V, E, n_0)$ where:

- V is a finite set of elements called **nodes** or vertices (with $|V| \geq 2$).
- $E \subseteq V \times V$ are the **edges**. An edge $e = (n, y) \in E$ is an outgoing edge of n and an incoming edge of y.
- $n_0 \in V$ is called the **root** of the tree. For every $n \in V \setminus \{n_0\}$, there is a unique path from n_0 to n. The root n_0 has no incoming edges.

- **Leaves (Z):** Nodes that do not have any outgoing edges. Denoted by $Z \subseteq V$.
- **Children ($C(n)$):** For a node $n \in V$, the children are $C(n) = \{y \in V \mid (n, y) \in E\}$.
- **Actions ($A(n)$):** For a node $n \in V$, the available actions are the outgoing edges $A(n) = \{e \in E \mid e = (n, y) \text{ for some } y \in V\}$.

2 Formal Definition of Extensive Form Games

Definition: Extensive Form Game (Perfect Information)

An extensive form game is a tuple $\Gamma = \langle N, V, E, n_0, (V_i)_{i \in N}, (u_i)_{i \in N} \rangle$ where:

- N is a finite set of players.
- (V, E, n_0) is a directed rooted tree.
- $(V_i)_{i \in N}$ is a partition of the non-leaf nodes (the nodes where decisions are made), i.e., a partition of the set $V \setminus Z$.
- $(u_i)_{i \in N}$ is a collection of utility functions, where $u_i : Z \to \mathbb{R}$ assigns a payoff to every leaf node for player i.

Let $J : V \setminus Z \to N$ be a function such that $J(n) = i$ if $n \in V_i$. The game proceeds in the following way:

1. Player $J(n_0)$ initiates the game by choosing a possible action from $A(n_0)$. Equivalently, he chooses a node $n_1 \in C(n_0)$.
2. If n_1 is a leaf, the game ends there. Otherwise, player $J(n_1)$ chooses a node $n_2 \in C(n_1)$.
3. In this way, the game continues until a leaf node is reached, at which point payoffs are distributed according to (u_i).

Note

Perfect Information: These games are perfect information games. The players know exactly what has happened before making their moves (they know exactly which node they are at).

3 Strategies in Extensive Form Games

Definition: Pure Strategy

A pure strategy for player i, denoted s_i, is a mapping which maps each node $n \in V_i$ (the nodes player i controls) to an element of $A(n)$ (an available action at that node). Equivalently, it maps n to a child node in $C(n)$.

$$s_i : V_i \to \bigcup_{n \in V_i} A(n)$$

such that $s_i(n) \in A(n)$ for all $n \in V_i$.

Crucial Note: A strategy must define an action for *every* node in V_i, even for nodes that might never be reached if the player follows that very strategy earlier in the tree. This is necessary to evaluate deviations and subgames.

Example: Strategy Example

If Player 1 controls nodes n_0 and n_4, a valid strategy s_1 must specify a choice at both nodes. For example:

$$s_1(n_0) = R \quad \text{and} \quad s_1(n_4) = A$$

This strategy is denoted as (R, A). Even if playing R at n_0 means node n_4 is never reached, the strategy still requires specifying action A at node n_4.